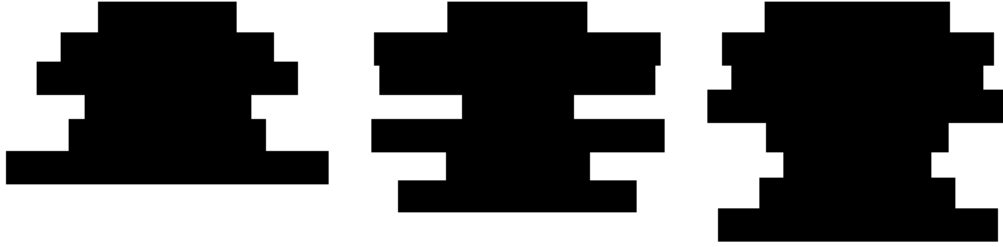

Latent-Space Forecasting of URANS Wake Flows Across Reynolds Numbers and LLM-Assisted Transition Regime Identification

Research assistance: GPT-5.2, EXAONE-4.01-32B
Inference models: gpt-4.1-mini, gemini-2.5-flash, claude-haiku-4-5



Abstract

Accurate prediction of unsteady wake flows across Reynolds numbers remains challenging due to high-dimensional spatiotemporal dynamics and regime-dependent behavior. This study presents a compact, data-driven framework built on time-resolved unsteady Reynolds-averaged Navier–Stokes (URANS) simulations for $Re \in \{1000, 2000, 3000, 4000, 5000, 7000, 9000, 11000, 22000\}$. Two-component velocity snapshots (u, v) are compressed using principal component analysis (PCA), yielding a low-dimensional latent representation with near-lossless reconstruction at $L = 50$. Using the $Re = 22000$ dataset, we formulate short-horizon latent forecasting from a finite history window and compare a linear extrapolation baseline with regularized linear predictors; Elastic Net provides the most consistent improvement and approaches the PCA reconstruction bound after decoding. Finally, as an auxiliary analysis, lightweight LLMs are queried as black-box similarity judges between endpoint references, and the entropy of repeated outputs is used as a heuristic uncertainty indicator for regime variation. Together, the results demonstrate an efficient pipeline for latent-space forecasting and a complementary, uncertainty-oriented view of transitional behavior from compressed URANS data. Notably, the LLM-based uncertainty trends consistently interpret the PCA-derived flow representations and provide an effective signal for transition-regime screening across Reynolds numbers.

1 Introduction

Turbulent flows exhibit high-dimensional, multiscale, and strongly nonlinear dynamics, which makes reliable prediction over long horizons challenging. While the Navier–Stokes equations provide a complete description, direct numerical simulation becomes prohibitively expensive at high Reynolds numbers (1). This motivates reduced-order representations that retain dominant flow structures while enabling efficient time evolution (2). A common approach is to project flow fields onto a low-dimensional subspace and model the reduced dynamics. Methods based on proper orthogonal decomposition (POD) and related operator-based formulations can compactly represent energetic

structures (3), but predictive performance may degrade in strongly nonlinear regimes due to truncation, instability, and model-form limitations (4). Data-driven approaches can learn reduced representations and temporal evolution from data, yet robust generalization across operating conditions remains difficult in chaotic flow systems (2).

In this study, we investigate reduced-order prediction of unsteady wake flows using time-resolved unsteady Reynolds-averaged Navier–Stokes (URANS) simulations. Two-component velocity fields are compressed via principal component analysis (PCA) into a low-dimensional latent space, where short-horizon forecasting is posed as a multivariate time-series problem and decoded back to the physical space for error evaluation (5; 6). The analysis spans multiple Reynolds numbers to examine how compression efficiency and near-term predictability vary as wake dynamics and turbulence-closure responses change (6; 7). Finally, lightweight large language models (LLMs) are used as black-box similarity judges on the reduced representations to provide an auxiliary, uncertainty-oriented view of regime variation, without claiming physical interpretability or replacing turbulence modeling.

2 Problem Formulation

We consider a two-dimensional URANS flow over a spatial domain $\Omega \subset \mathbb{R}^2$, with $\mathbf{x} \in \Omega$. After spatial discretization on N_p grid points, each snapshot is represented by the two-component velocity field $(u(\mathbf{x}, t), v(\mathbf{x}, t))$. Here, u and v correspond to the streamwise and transverse velocity components, respectively (equivalently $u \equiv u_1$ and $v \equiv u_2$ in index notation). By stacking the two components, the high-dimensional state vector is written as

$$\mathbf{u}(t) = [u(\mathbf{x}_1, t), \dots, u(\mathbf{x}_{N_p}, t), v(\mathbf{x}_1, t), \dots, v(\mathbf{x}_{N_p}, t)]^\top \in \mathbb{R}^{2N_p}.$$

Here, $\mathbf{u}(t)$ denotes the resolved mean-flow variables obtained from URANS, while the effects of turbulent fluctuations are modeled through a turbulence closure. Although URANS significantly reduces the computational cost compared to direct numerical simulation, the resulting system remains high-dimensional and nonlinear, and its temporal evolution can exhibit complex dynamics depending on the flow configuration and Reynolds number.

To enable predictive modeling in a reduced space, we assume that the high-dimensional URANS state vectors $\mathbf{u}(t)$ admit an accurate approximation in a low-dimensional subspace of $\mathbb{R}^{2N_p}$. Accordingly, we introduce a generic encoding–decoding mapping,

$$\mathbf{z}(t) = \mathcal{E}(\mathbf{u}(t)), \quad \hat{\mathbf{u}}(t) = \mathcal{D}(\mathbf{z}(t)), \quad (1)$$

where $\mathcal{E} : \mathbb{R}^{2N_p} \rightarrow \mathbb{R}^r$ and $\mathcal{D} : \mathbb{R}^r \rightarrow \mathbb{R}^{2N_p}$ denote an encoding and decoding operator, respectively, and $r \ll 2N_p$ is the latent dimension. This formulation is agnostic to the specific choice of reduced representation and is introduced here to define the general latent-space prediction framework.

The temporal evolution of the flow is modeled in the latent space as a discrete-time dynamical system. Let $\{\mathbf{z}_n\}_{n=1}^T$ denote the latent states sampled at uniform time intervals Δt . We define a latent dynamics operator $\mathcal{F}$ such that

$$\mathbf{z}_{n+1} = \mathcal{F}(\mathbf{z}_{n-k+1}, \dots, \mathbf{z}_n), \quad (2)$$

where $k \geq 1$ denotes the length of the temporal history used to predict the next state. This formulation allows the latent dynamics to incorporate memory effects and nonlinear temporal dependencies that arise from the coupled mean-flow evolution and turbulence modeling, without assuming Markovian behavior in the reduced space.

Given an initial sequence of latent states $\{\mathbf{z}_1, \dots, \mathbf{z}_k\}$, the model generates a predicted latent trajectory $\{\hat{\mathbf{z}}_{k+1}, \hat{\mathbf{z}}_{k+2}, \dots\}$ by iteratively applying $\mathcal{F}$. The corresponding high-dimensional flow prediction at time step n is then obtained via the decoder as

$$\hat{\mathbf{u}}_n = \mathcal{D}(\hat{\mathbf{z}}_n). \quad (3)$$

In this study, the encoder–decoder mapping is instantiated using PCA, yielding a linear and interpretable latent representation of the URANS flow fields. The latent dynamics operator $\mathcal{F}$ is implemented using regularized linear regression models, with Elastic Net (8) adopted as the primary

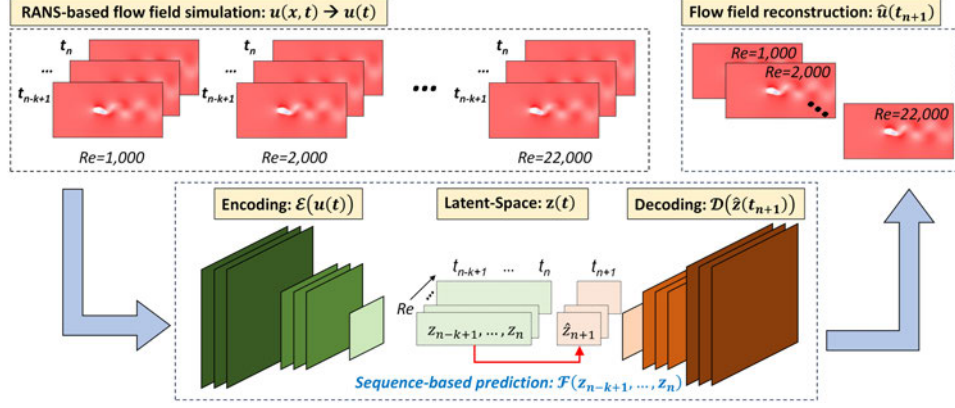


Figure 1: Overall workflow of the latent-space dynamical prediction framework.

predictor and Ridge (9) and Lasso (10) regressions employed as reference baselines for comparative evaluation.

Finally, we introduce a transition-regime screening task based on PCA-derived latent representations. Each Reynolds-number case is summarized as a fixed latent flow image, and intermediate cases are compared against low- and high-Re endpoint references using a black-box similarity judge that returns a soft preference. By repeating the comparison and tracking the consistency of the returned preference, we obtain an uncertainty signal. Reynolds numbers exhibiting higher uncertainty are treated as candidates for transitional behavior.

3 Methodology

In the present study, the encoding–decoding operators $\mathcal{E}$ and $\mathcal{D}$ are instantiated using PCA, yielding a linear and interpretable latent representation. Figure 1 provides an overview of the proposed methodology for turbulent flow prediction based on latent-space dynamical learning. High-dimensional flow fields are first obtained from time-dependent URANS simulations and used as input data. These flow snapshots are mapped onto a low-dimensional latent representation through an encoding process. The temporal evolution of the latent variables is then modeled as a sequence-based prediction problem, where future latent states are inferred from a finite history of past states. Finally, the predicted latent states are decoded back into the physical space to reconstruct the flow field at a target time, enabling assessment of both in-distribution prediction accuracy and generalization across Reynolds numbers. The image-based regime identification using large language models is introduced separately in Section 3.4 as a complementary analysis.

3.1 URANS-Based Flow Simulation and Data Preparation

Time-dependent flow fields were generated using two-dimensional URANS simulations, which describe the conservation of mass and momentum for the resolved mean flow,

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (4)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} - \frac{\partial}{\partial x_j} \overline{u'_i u'_j}, \quad (5)$$

where indices $i, j \in \{1, 2\}$ denote the streamwise and transverse directions, respectively, and the Einstein summation convention is implied (11). Here, $\bar{u}_i$ and $\bar{p}$ are the Reynolds-averaged velocity components and pressure, and $\overline{u'_i u'_j}$ is the Reynolds stress tensor, which is closed using a turbulence model. In the present two-dimensional configuration, the mean velocity field is written as $\bar{\mathbf{u}} = (\bar{u}, \bar{v})$.

URANS simulations were performed for a selected set of Reynolds numbers, $\text{Re} \in \{1000, 3000, 5000, 7000, 9000, 11000, 13000, 15000, 22000\}$, covering a broad range from low to high Reynolds numbers. This set provides sufficient variation in wake dynamics while remaining

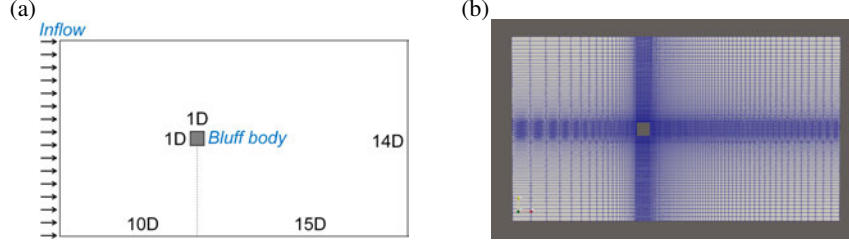


Figure 2: Schematic illustration of (a) geometrical domain and (b) computational mesh configuration

computationally tractable for long, time-resolved simulations. Across this range, the temporal evolution of the mean flow is explicitly resolved, while the effects of turbulent fluctuations are modeled through a turbulence closure (12).

The flow configuration consists of a canonical two-dimensional bluff-body wake (Fig. 2). A square prism of size $1D \times 1D$ is placed in a rectangular domain normalized by the obstacle diameter D , with the inlet located $10D$ upstream and the outlet $15D$ downstream of the prism. The domain height is $14D$. A uniform inflow velocity U_∞ is prescribed at the inlet, with standard outflow and lateral boundary conditions applied to minimize confinement effects.

All simulations were conducted using the finite-volume solver OpenFOAM with a transient URANS formulation and a $k-\omega$ turbulence model (13). A consistent solver and turbulence-closure setup was used for all Reynolds numbers to ensure comparability across cases. After initial transients decayed and a statistically stationary state was reached, snapshots of the Reynolds-averaged velocity field $(\bar{u}, \bar{v})$ were sampled at uniform temporal intervals. For brevity, we denote $(\bar{u}, \bar{v})$ by (u, v) in the remainder of the paper. Each snapshot was vectorized to form the high-dimensional state representation used for the subsequent latent-space analysis. At lower Reynolds numbers, the modeled eddy-viscosity contribution becomes small and the URANS solutions approach laminar-like behavior (14).

3.2 Latent Space Construction and Temporal Dynamics Prediction

Following the general formulation introduced in Section 2, the encoder-decoder mapping is instantiated using PCA, yielding a linear and interpretable latent representation of the URANS velocity fields. Each flow snapshot, consisting of the two-component velocity field (u, v) , is projected onto a low-dimensional latent vector $\mathbf{z}(t) \in \mathbb{R}^r$, with $r \ll N$, while retaining the dominant spatiotemporal wake structures. The temporal evolution of the flow is modeled directly in the latent space. Latent states sampled at uniform time intervals are treated as a multivariate time series, and short-horizon prediction is formulated using a finite history window of length k . The next latent state is predicted from the previous k states using a regression-based latent dynamics model. We consider multiple regression models for this latent predictor and compare their short-horizon performance. This separation between spatial compression and temporal prediction enables efficient reduced-order modeling of unsteady wake dynamics and facilitates systematic evaluation of predictive accuracy and generalization behavior across Reynolds numbers.

3.3 Flow Field Reconstruction and Evaluation Metrics

Following the latent-space prediction described in Section 2, the predicted latent states are decoded back into the physical space to reconstruct the corresponding velocity fields. This reconstruction enables direct comparison between predicted flow fields and the reference URANS solutions, allowing quantitative assessment of prediction accuracy in the original high-dimensional state space. Reconstruction accuracy is first evaluated using a global normalized ℓ_2 error defined at each time step as $\varepsilon(t) = \|\hat{\mathbf{u}}(t) - \mathbf{u}(t)\|_2 / \|\mathbf{u}(t)\|_2$ which provides a compact measure of instantaneous prediction fidelity. In addition, spatial distributions of reconstruction error are examined to identify localized discrepancies associated with wake structures, shear layers, and regions of strong velocity gradients.

Beyond instantaneous errors, statistical flow characteristics are used to assess the long-term predictive behavior of the latent-space model. Time-averaged velocity fields and selected second-order statistics are computed from the reconstructed trajectories and compared against reference URANS statistics.

These comparisons assess whether the latent-space predictions preserve the dominant mean-flow organization and variability of the wake. All evaluation metrics are applied to both in-distribution cases, where the Reynolds number is included in the training data, and out-of-distribution cases, where predictions are performed at Reynolds numbers not seen during training. This distinction enables systematic analysis of prediction accuracy, error accumulation, and generalization behavior as the flow regime varies.

3.4 Identification of transition regime via Image-Based Inference

In addition to latent-space dynamical prediction, this study explores the use of large language models (LLMs) as a novel inference tool for identifying transitional flow regimes from reduced flow-field images. Figure 3 illustrates the overall workflow of the proposed LLM-based image inference procedure, from reduced flow-field representations at reference Reynolds numbers to entropy-based identification of transition regimes. Unlike conventional classifiers trained on fixed labels or regression targets, LLMs are used here as black-box visual similarity judges: given endpoint reference images, they assign soft preferences to intermediate cases. We summarize the variability of these preferences via repeated queries and entropy to obtain an uncertainty proxy for regime change. To reduce model-specific bias and assess robustness, multiple lightweight LLMs are employed in this study, including gpt-4.1-mini, gemini-2.5-flash, and claude-haiku-4-5.

Reduced flow-field images corresponding to two reference Reynolds numbers, $Re = 1000$ and $Re = 22000$, are first provided to the LLMs as endpoint examples representing low- and high-Reynolds-number wake regimes, respectively. Given an input image at an intermediate Reynolds number, each LLM is prompted to infer its association with either reference regime based solely on the visual characteristics of the reduced flow representation. The inference outcome is encoded as a binary preference vector, reflecting the model’s relative association with each endpoint regime.

To ensure robustness and mitigate stochastic variability inherent to LLM-based inference, the classification process is repeated over multiple independent realizations. For each Reynolds number, ten independent inference trials are conducted, and the resulting one-hot vectors (15; 16) are averaged to obtain a probabilistic measure of regime association. The entropy of the averaged output distribution is then computed and used as a quantitative indicator of classification uncertainty. High entropy values indicate Reynolds numbers for which the LLMs exhibit difficulty in decisively assigning the flow to either reference regime, suggesting the presence of a transitional or transient flow regime.

The use of LLM-based image inference in this context departs from conventional turbulence analysis and data-driven classification approaches. The LLM outputs are treated strictly as heuristic judgments of image-level pattern similarity, without any claim of physical interpretability or causal modeling. The analysis focuses on uncertainty trends induced by repeated inference across Reynolds numbers, enabling identification of Reynolds-number ranges associated with increased regime ambiguity. This perspective provides a complementary, uncertainty-oriented view of flow regime variation that augments the latent-space dynamical framework.

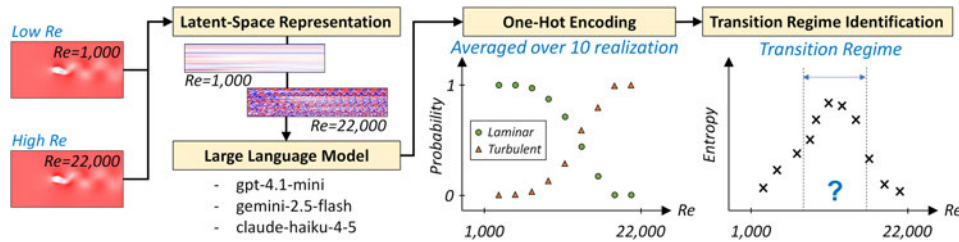


Figure 3: LLM-based identification of transitional flow regimes using reduced flow-field images.

4 Results

4.1 PCA-Based Compression of Time-Resolved Velocity Fields

We first construct a compact representation of the time-resolved velocity fields using PCA. For each Re in $\{1000, 2000, 3000, 4000, 5000, 7000, 9000, 11000, 22000\}$, we collect the two-component

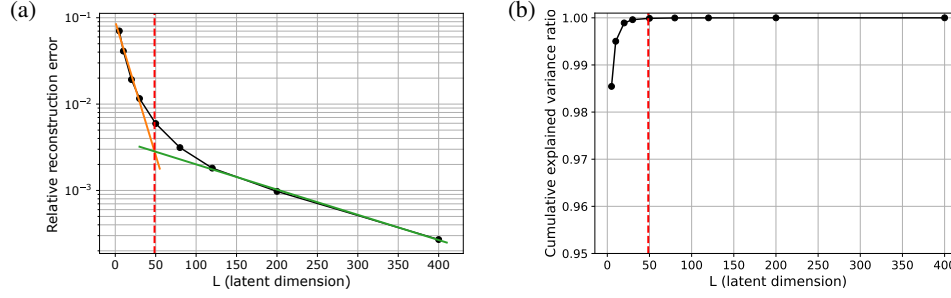


Figure 4: PCA compression performance versus latent dimension L . (a) Relative reconstruction error (normalized ℓ_2 error) showing a two-regime decay; the fitted trends intersect at $L^* \approx 48.56$ (vertical dashed line). (b) Cumulative explained variance ratio, approaching unity near $L \approx 50$.

velocity snapshots (u, v) at a time resolution of 0.01 from $t = 15$ s to $t = 20$ s, yielding 501 time steps per case. Each snapshot is defined on a grid with 15000 pixels, so the raw data tensor has shape $(N_{\text{Re}}, N_t, N_p, N_f) = (9, 501, 15000, 2)$. We then vary the PCA latent dimension $L \in \{5, 10, 20, 30, 50, 80, 120, 200, 400\}$ and evaluate reconstruction quality using the relative reconstruction error (normalized ℓ_2 norm), together with the cumulative explained variance ratio (Fig. 4). The reconstruction error exhibits a clear two-regime decay: it decreases rapidly for small L and more slowly for large L . To quantify this behavior, we fit two trends in the small- L and large- L ranges and define their intersection as the optimal compression dimension. The intersection occurs at $L^* \approx 48.5569$, where the reconstruction error is 0.0028343 and the cumulative variance ratio is 0.999875, indicating near-lossless reconstruction. Based on this result, we select $L = 50$ as the practical latent dimension that guarantees a similarly high reconstruction fidelity. Consequently, the PCA-compressed dataset has shape $(9, 501, 50)$, meaning that each (u, v) field of dimension 15000×2 is reduced to a 50-dimensional latent vector. Finally, we apply z -score normalization over the full dataset. This representation can be used directly as a time series in latent space, or equivalently interpreted as nine images of size $(501, 50)$, one per Re (Fig. 5).

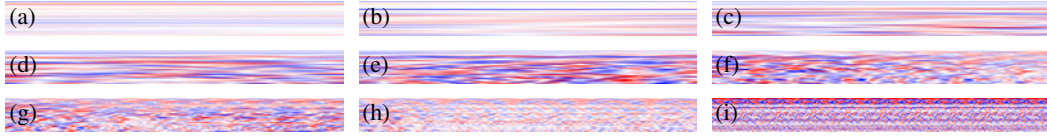


Figure 5: z -score normalized PCA latent images of size $(501, 50)$ for each Reynolds number: (a) Re = 1000, (b) 2000, (c) 3000, (d) 4000, (e) 5000, (f) 7000, (g) 9000, (h) 11000, and (i) 22000.

4.2 Short-Horizon Forecasting in the Turbulent Regime

Next, we focus on a reference turbulent case ($\text{Re} = 22000$) and investigate whether the reduced-order representation enables accurate short-horizon prediction. We use the PCA-compressed and normalized data in the $(501, 50)$ form with a 0.01 time interval. Our goal is to predict the next latent vector from a short history window of length 20, i.e., using the previous 20 time steps as input to estimate the immediately following step. We construct the dataset by sliding a one-step window across time, producing 481 input-output pairs. Among them, the first 400 pairs are used for training and the remaining 81 pairs (corresponding to the later time range, roughly $t = 19.2$ s to $t = 20$ s) are used for testing. When validation is required, we reserve 10% of the training set as a validation split. We compare four models: (i) a one-step linear extrapolation baseline (used as a reference), (ii) Ridge regression, (iii) Lasso regression, and (iv) Elastic Net regression. All models are trained in the latent space and then evaluated by predicting the test interval. The results (Fig. 6) show that the regression-based models (ii–iv) substantially outperform the baseline (i). Among them, Elastic Net provides the most consistent performance. Importantly, after decoding the predicted latent vectors back to the original (u, v) fields, Elastic Net achieves a reconstruction quality that is nearly indistinguishable from the intrinsic PCA decoding fidelity (Fig. 7). This indicates that the dominant prediction error is limited by the PCA reconstruction bound rather than the forecasting model itself.

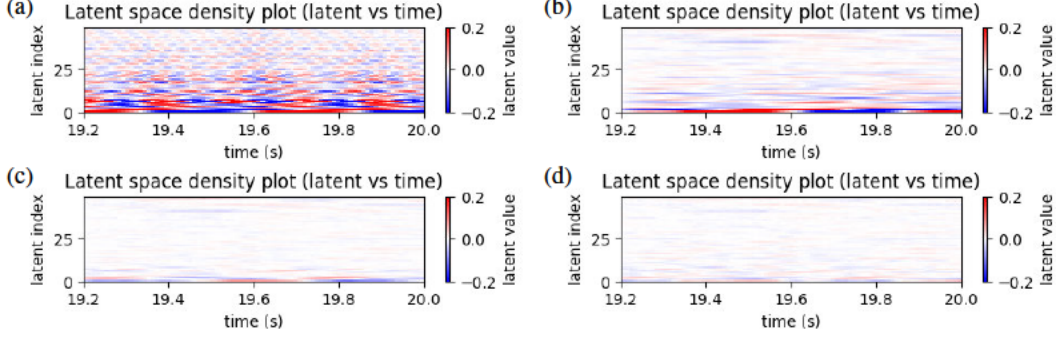


Figure 6: Test-set latent-space error maps (model prediction minus PCA-decoded ground truth) at $Re = 22000$ for (a) linear extrapolation, (b) Ridge, (c) Lasso, and (d) Elastic Net.

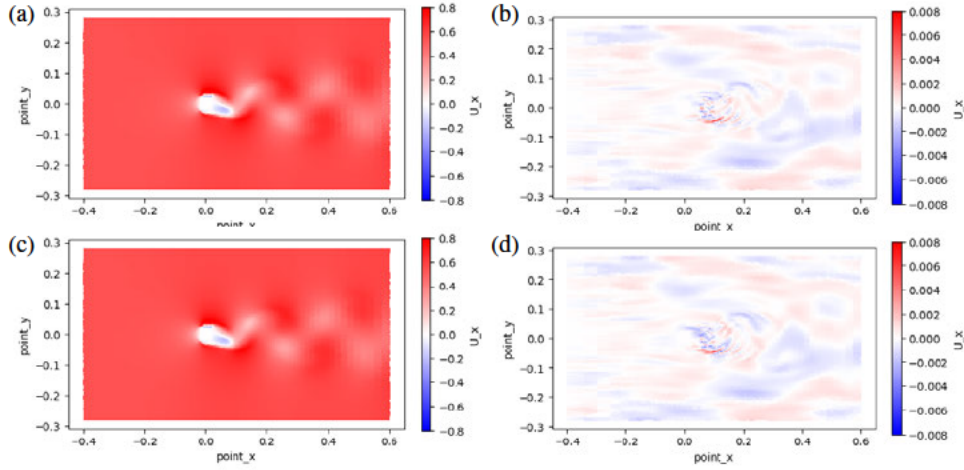


Figure 7: Comparison of inverse-PCA reconstructions and Elastic Net short-horizon prediction for the streamwise velocity field u_x at $Re = 22000$. (a) u_x reconstructed by decoding the PCA-compressed representation (inverse PCA). (b) Reconstruction error, $u_x^{PCA \text{ decode}} - u_x^{orig}$. (c) Elastic Net one-step prediction in latent space decoded back to the physical field (inverse PCA). (d) Prediction error, $u_x^{EN \text{ pred}} - u_x^{orig}$, showing that the Elastic Net forecast closely matches the original field.

Furthermore, the same latent forecasting pipeline was also applied to the remaining eight Re cases, and we verified that the prediction performance remains consistently strong across all Re .

4.3 LLM-Based Identification of the transition regime

Finally, we propose an LLM-based transition-regime indicator using the PCA flow images described in Fig. 5. The goal is to extract a heuristic uncertainty trend across Reynolds numbers from reduced URANS representations. We consider three lightweight models, gpt-4.1-mini, gemini-2.5-flash, and claude-haiku-4-5. Each model is provided with two endpoint reference images via a consistent JSON prompt: a low- Re reference at $Re = 1000$ with $(y_1, y_2) = (1, 0)$ and a high- Re reference at $Re = 22000$ with $(y_1, y_2) = (0, 1)$. Here, y_1 and y_2 denote the model’s soft preference for the low- and high- Re references, respectively. At $Re = 1000$, the modeled eddy-viscosity contribution is small and the URANS solution is laminar-like; at $Re = 22000$, the wake is fully unsteady and turbulence-model effects are significant.

We then query the remaining Reynolds-number images one by one. For each Re , we run $K = 10$ realizations per LLM (temperature 0.2) and average the predicted probabilities (y_1, y_2) , along with the corresponding binary entropy $H(y_1, y_2)$ to quantify uncertainty. Figure 8(a–c) reports the mean and standard deviation of $y_1(Re)$ and $y_2(Re)$, and we define the maximally confusing intersection Re^* as the point where $y_2(Re) = 0.5$. Figure 8(d–f) shows $H(Re)$, from which we infer the transition

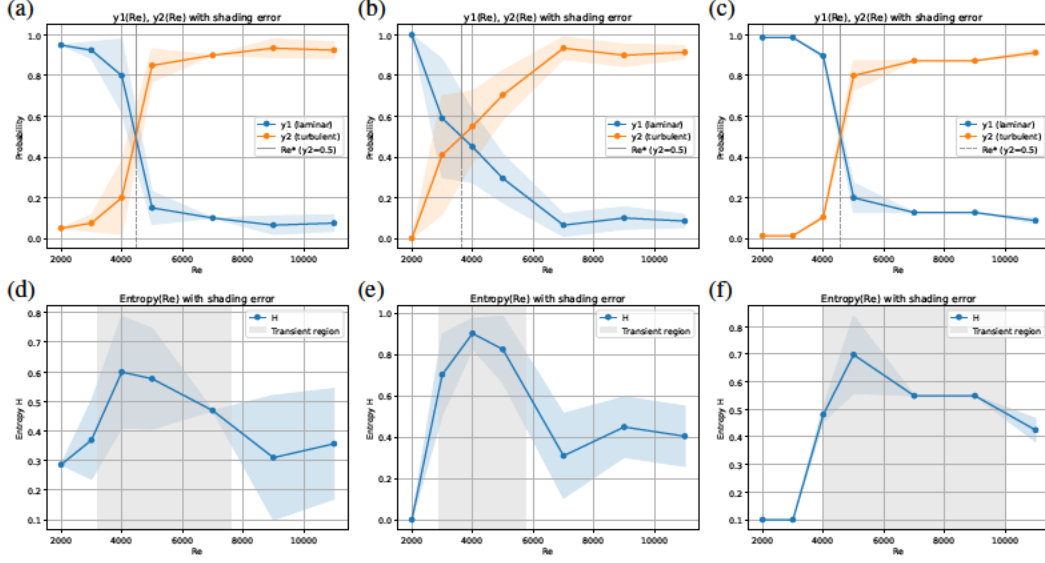


Figure 8: LLM-based identification of the transition regime in Re . Panels (a–c) show one-hot probabilities $y_1(Re)$ (laminar) and $y_2(Re)$ (turbulent), with the maximally confusing intersection Re^* given by $y_2 = 0.5$ (dashed line). Panels (d–f) show the entropy $H(Re)$, used to define the transition regime (shaded). Models: (a,d) gpt-4.1-mini, (b,e) gemini-2.5-flash, (c,f) claude-haiku-4-5.

regime (Re_{min}, Re_{max}) by selecting the entropy maximum and including neighboring Reynolds numbers until $H(Re)$ falls below 70% of its peak value.

As an external physical reference, we also analyze macroscopic aerodynamic indicators, namely C_L and C_D , as functions of Re (Fig. 9). From these trends, transitional behavior is observed over approximately $Re = 3000$ to 8000 . Table 1 summarizes the LLM-inferred Re^* and (Re_{min}, Re_{max}), and additionally reports overlap metrics (IoU, precision, recall) computed relative to the baseline interval $Re = 3000$ – 8000 . To quantify agreement, IoU measures the fractional overlap between the predicted and baseline intervals, precision measures the fraction of the predicted interval that lies within the baseline, and recall measures the fraction of the baseline interval covered by the prediction. Among the three models, gpt-4.1-mini shows the strongest agreement with the baseline (IoU = 0.881) and perfect precision (1.000), and high recall (0.881), indicating that its inferred transition range is fully contained within $Re = 3000$ – 8000 while covering 88.1% of that baseline range. In contrast, gemini-2.5-flash yields a lower IoU (0.539) with high precision (0.965) but modest recall (0.550), suggesting a conservative (narrow) transition estimate that largely stays within the baseline but captures only about half of it. Finally, claude-haiku-4-5 attains IoU = 0.569

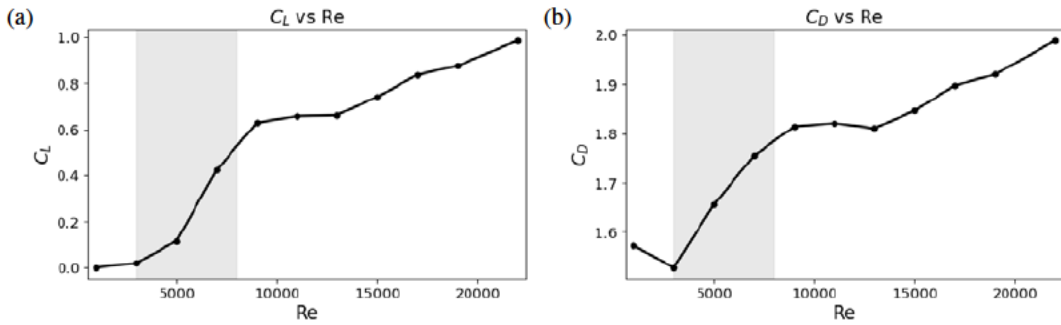


Figure 9: Macroscopic aerodynamic coefficients used as a physical baseline for validating the LLM-based transitional-regime estimate. Shown are (a) lift coefficient C_L and (b) drag coefficient C_D computed from the original (u, v) fields as functions of Re . The observed trend change suggests a transitional range of approximately $Re = 3000$ – 8000 (shaded).

Table 1: Summary of the LLM-inferred intersection Re^* and transition regime (Re_{min}, Re_{max}), along with overlap metrics (IoU, precision, recall) relative to the baseline $Re = 3000\text{--}8000$.

Model	Re^*	transition regime		IoU	Precision	Recall
		Re_{min}	Re_{min}			
gpt-4.1-mini	4462	3218	7621	0.881	1.000	0.881
gemini-2.5-flash	3643	2899	5750	0.539	0.965	0.550
claude-haiku-4-5	4569	4035	9974	0.569	0.668	0.793
Baseline (from C_L/C_D analysis)	-	3000	8000	-	-	-

with recall 0.793 but reduced precision (0.668), reflecting a broader transition estimate that covers most of the baseline while extending beyond it toward higher Re . Overall, the inferred ranges are broadly consistent with the aerodynamic baseline, indicating that lightweight LLMs can provide a complementary uncertainty-oriented signal for identifying the transition regime from reduced URANS flow representations.

5 Discussion

We presented a data-driven framework for turbulent-flow modeling that unifies three capabilities in a common reduced representation: compact encoding of time-dependent URANS velocity fields, short-horizon forecasting in latent space, and uncertainty-oriented identification of the transition regime. The key observation is that PCA yields an information-preserving and standardized spatiotemporal descriptor of the wake dynamics, enabling both learning and cross- Re comparison at a fraction of the original dimensional cost. In this latent space, simple regularized predictors already capture the dominant short-time evolution, and the remaining discrepancy after decoding is largely constrained by the intrinsic compression–reconstruction fidelity rather than the forecasting model itself. This suggests that, for near-term prediction, representation quality is the primary bottleneck. Finally, we showed that lightweight LLMs can meaningfully interpret PCA-derived flow images to produce a coherent uncertainty signal across Reynolds numbers. By anchoring laminar and turbulent endpoints and tracking maximum confusion and entropy peaks, the inferred transition regime aligns with macroscopic aerodynamic trends in C_L and C_D (17), indicating that compact spatiotemporal flow descriptors can serve as an effective interface between reduced-order CFD outputs and foundation-model-based regime screening. The results presented here should be interpreted in the context of the modeling choices and scope adopted in this study. The latent representation is based on PCA, which is linear and may not fully capture nonlinear structures in more complex or higher-dimensional flows, despite achieving near-lossless reconstruction for the present wake configurations. The latent dynamics are evaluated in a short-horizon, one-step prediction setting; long-horizon stability and error accumulation are not addressed. In addition, all results are based on URANS simulations, and the learned dynamics may inherit biases associated with the turbulence-closure model, particularly in transitional regimes. The LLM-based regime identification is intentionally heuristic and relies on uncertainty trends from image-level similarity rather than physical classification. Future work will focus on extending the framework to nonlinear encoders, longer-horizon latent forecasting, and higher-fidelity or three-dimensional flow data. The LLM-based analysis could also be complemented by physics-informed or data-driven uncertainty measures.

6 Conclusion

We introduced a compact, data-driven workflow for analyzing time-dependent wake flows across Reynolds numbers by unifying three components: linear compression, latent-space forecasting, and LLM-based regime inference. By representing high-dimensional (u, v) fields as PCA-derived spatiotemporal descriptors, we obtain a standardized, information-preserving input that can be directly consumed by lightweight predictive models and, importantly, by modern LLMs. Our results demonstrate that LLMs can meaningfully interpret these reduced flow representations to produce an uncertainty-oriented signal that supports transition-regime localization, complementing conventional aerodynamic indicators. This direction suggests a scalable pathway toward automated, physics-aware screening and monitoring of complex flow behavior, with potential impact on high-throughput CFD analysis, surrogate modeling, and digital-twin pipelines where rapid decision support is required.

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